

CHAPTER THREE

MEASURES OF CENTRAL TENDENCY AND DISPERSION

In this section we discuss two important aspects of data which are its **center** and its **spread**. The *mean*, *median*, and the *mode* are measures of central tendency that describe **where data are centered**. The *range*, *variance*, and *standard deviation* are measures of dispersion that describe the spread of data.

3.1. Measures of Central Tendency: *Mean, Mode, and Median*

In many real-life situations, it is helpful to describe data by a single number that is most representative of the entire collection of numbers. Such a number is called a **measure of central tendency**. The most commonly used measures are as follows.

1. The **mean**, or **average**, of N numbers is the sum of the numbers divided by N.
2. The **median** of N numbers is the middle number when the numbers are written in order. If N is even, the median is the average of the two middle numbers.
3. The **mode** of N numbers is the number that *occurs most frequently*. If two numbers tie for most frequent occurrence, the collection has two modes and is called **bimodal**.

➤ **Mode:** It is defined as the value which occurs with the highest frequency in the data. Thus, it is used when one is interested in the most common value in a distribution. A mode may not exist and usually this happens when no data value occurs more frequently than all the others. Also a mode may not be unique.

Example 3.1 The final grades of a class of six graduate students were A, C, B, C, A, B.

What is the mode?

Solution: Since the grades occur at the same frequency then there is no mode for this data.

Example 3.2 A sample of the records of motor vehicle bureau shows that 18 drivers in a certain age group received 3, 2, 0, 0, 2, 3, 3, 1, 0, 1, 0, 3, 4, 0, 3, 2, 3, 0 traffic tickets during the last three years.

Find the mode?

Solution: The mode consists of 0 and 3 since they occur six times on the list.

➤ **Median:** Another measure of the center is the median. The median usually is found when we have an ordered distribution. It is computed as follows. We arrange the numerical data in an ascending or descending order (from smallest to largest or vice versa). If n denotes the size of the set of data then the median can be found by using the median rank(MR); $MR = \frac{n+1}{2}$

If MR is a whole number then the median is the value in that position. If MR ends in .5, we take the sum of the adjacent positions and divide by 2. Unlike the mode, the median always exists and is unique. But it may or may not be one of the given data values. Note that extreme values (smallest or largest) do not affect the median.

Example 3.3: Among groups of 40 students interviewed at each of 10 different colleges, 18, 13, 15, 12, 8, 3, 7, 14, 16, 3 said that they jog regularly. *Find the median.*

Solution: First, arrange the numbers from smallest to largest to obtain 3 3 7 8 12 13 14 15 16 18. Next, compute the MR = (10+1)/ 2 = 5.5. Hence, the median is (12+13)/2 = **12.5**

Example 3.4

Nine corporations reported that in 1982 they made cash donations to 9, 16, 11, 10, 13, 12, 6, 9, and 12 colleges. Find the **median number**.

Solution: Arranging the numbers from smallest to largest to obtain 6 9 9 10 11 12 12 13 16

The median rank is MR = (9+1)/ 2 = 5. The median is 11.

➤ **Mean** - commonly referred to as the average or arithmetic mean.

- most widely used measure of central location.

$$\bar{X} = \frac{\text{Sum of all values in the data set}}{\text{Total number of observations}} = \frac{\sum x}{n}$$

➤ Types of mean

a) Simple arithmetic mean

Definition: If $X_1, X_2, \dots, X_N$ are the values of a variable X, then arithmetic mean of X denoted by $\bar{X}$ is defined as

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_N}{N} = \frac{\sum_{i=1}^N X_i}{N} = \frac{(\sum X)}{N}$$

For the case of discrete grouped data, if X assumes k distinct values $X_1, X_2, \dots, X_k$ with respective frequencies

$f_1, f_2, \dots, f_k$, then $\bar{X}$ (the Arithmetic mean of X) is

$$\bar{X} = \frac{\sum_{i=1}^k f_i X_i}{N}, \text{ where } N = \sum_{i=1}^k f_i$$

For the case of a frequency distribution of a continuous variable grouped in to class-intervals, if there are k classes with respect to mid points $X_1, X_2, \dots, X_k$ and respective frequencies $f_1, f_2, \dots, f_k$ then

$$\text{Arithmetic mean } \bar{X} \text{ of } X \text{ is } \bar{X} = \frac{\sum_{i=1}^k f_i X_i}{N}, \text{ where } N = \sum_{i=1}^k f_i$$

Example3.1: Compute the Arithmetic mean for the following frequency distributions.

a)

Values (X_i)	20	21	22	23	24	25
Frequencies(f_i)	5	5	7	6	6	7

b)

Marks (X)	f_i
10-19	4
20-29	5
30-39	8
40-49	6
50-59	2

Mathematical Properties of the arithmetic mean

Arithmetic mean possesses some very interesting and important mathematical properties as given below.

1. If we subtract an arbitrary constant from each of the observations the mean is also reduced by the constant value.

Example3.2: The mean of the observations 2, 4, 6, and 8 is

$$\frac{2+4+6+8}{4} = \frac{20}{4} = 5$$

. If we subtract from each of these observations 2, we get the new observation 0, 2, 4 and 6. The mean of the new observations is

$$\frac{0+2+4+6}{4} = 3$$

, which is mean of the original data minus the constant 2, i.e. $5-2=3$.

2. If we add an arbitrary constant to each of the observations the mean is also increased by the same constant value.
3. If we divide each observation of a set by an arbitrary constant the mean is reduced as many times as the constant divisor.

Example 3.3: If we have a set of data 2, 4, 6 and 8 then their mean is **5**. If we divide the given set of

data by 2, we get a new set of data, 1, 2, 3 and 4. The mean of the new set of data is $\frac{1+2+3+4}{4} = 2.5$

which is equal to $\frac{5}{2} = 2.5$

4. If a wrong figure has been used when computing the mean $(\bar{X})$ the correct mean can be obtained without repeating the whole process using the relation

Correct mean = wrong mean + $\frac{(\text{correct value} - \text{wrong value})}{N}$, where N is total number of observation

Example 3.4: The arithmetic mean of 20 observations is 20. But while calculating this, an observation 13 was misread as 30. Compute the correct mean.

Solution:

Given $N = 20, \bar{X} = 20$.

$$\begin{aligned} \text{Then } \bar{X}(\text{correct}) &= \bar{X}(\text{wrong}) + \frac{\text{Correct value} - \text{wrong value}}{N} \\ &= 20 + \frac{13 - 30}{20} \\ &= 20 - 0.85 \\ &= 19.15 \end{aligned}$$

b) Weighted mean

In the computation of arithmetic mean we assume that all items are of equal importance. It may not be so. Importance of different items can be shown by attaching suitable weights to them relative to their importance. If $W_1, W_2, \dots, W_N$ are the weights assigned to the values $X_1, X_2, \dots, X_N$ respectively, then the weighted mean is given as:

$$\bar{X}_w = \frac{W_1X_1 + W_2X_2 + \dots + W_NX_N}{W_1 + W_2 + \dots + W_N} = \frac{\sum_{i=1}^N W_i X_i}{\sum_{i=1}^N W_i}$$

(When $W_i=1$ for all $i=1,2,\dots,N$ then the mean becomes the arithmetic)

Example 3.5: A student was registered for five courses with 4,4,3,2 and 3 credit hours. She obtained B, A, C D and F grades respectively. The grading system is of the form A= 4, B=3, C=2, D=1 and F=0. Find the GPA of the student.

✓ *Which of the three measures of central tendency is the most representative?*

The answer is that it depends on the distribution of the data *and* the way in which you plan to use the data.

Which measure of central tendency is the most representative of the data shown in each frequency distribution?

a. Number	Tally	b. Number	Tally	c. Number	Tally
1	7	1	9	1	6
2	20	2	8	2	1
3	15	3	7	3	2
4	11	4	6	4	3
5	8	5	5	5	5
6	3	6	6	6	5
7	2	7	7	7	4
8	0	8	8	8	3
9	15	9	9	9	0

Solution:

- For this data, the mean is 4.23, the median is 3, and the mode is 2. Of these, the mode is probably the most representative.
- For this data, the mean and median are each 5 and the modes are 1 and 9 (the distribution is bimodal). Of these, the mean or median is the most representative.
- For this data, the mean is 4.59, the median is 5, and the mode is 1. Of these, the mean or median is the most representative.

Example 2 _ Comparing Measures of Central Tendency

On an interview for a job, the interviewer tells you that the average annual income of the company's 25 employees is \$60,849. The actual annual incomes of the 25 employees are shown below. What are the mean, median, and mode of the incomes? Was the person telling you the truth?

\$17,305, \$478,320, \$45,678, \$18,980, \$17,408,
 \$25,676, \$28,906, \$12,500, \$24,540, \$33,450,

\$12,500,	\$33,855,	\$37,450,	\$20,432,	\$28,956,
\$34,983,	\$36,540,	\$250,921,	\$36,853,	\$16,430,
\$32,654,	\$98,213,	\$48,980,	\$94,024,	\$35,671

Solution

$$\begin{aligned}
 \text{The mean of the incomes is} &= \frac{17,305 + 478,320 + 45,678 + 18,980 + \dots + 35,671}{25} \\
 &= \frac{1,521,225}{25} \\
 &= \underline{\underline{\$60,849}}.
 \end{aligned}$$

To find the median, order the incomes as follows.

\$12,500, \$12,500, \$16,430, \$17,305, \$17,408, \$18,980, \$20,432, \$24,540, \$25,676, \$28,906, \$28,956, \$32,654, \$33,450, \$33,855, \$34,983, \$35,671, \$36,540, \$36,853, \$37,450, \$45,678, \$48,980, \$94,024, \$98,213, \$250,921, \$478,320

From this list, you can see that the median (the middle number) is \$33,450. From the same list, you can see that \$12,500 is the only income that occurs more than once. So, the mode is \$12,500. Technically, the person was telling the truth because the average is (generally) defined to be the mean. However, of the three measures of central tendency *Mean*: \$60,849 *Median*: \$33,450 *Mode*: \$12,500 it seems clear that the median is most representative. The mean is inflated by the two highest salaries.

Advantages of the mean:

- It takes into account all observations
- It can be used for further statistical calculations and mathematical manipulations.

Disadvantages of the mean:

- ✓ Easily affected by extreme values.
- ✓ Cannot be computed if there are missing values due to omission or non-response
- ✓ In grouped data with open-ended class intervals, the mean cannot be computed.

Advantages of the MEDIAN:

- ✓ Not affected by extreme values.
- ✓ Can be computed even for grouped data with open ended class intervals.

Disadvantages of the MEDIAN:

- ✓ Observations from different data sets have to be merged to obtain a new median, whether group or ungrouped data are involved.

Advantage of the MODE:

- ✓ Can be easily identified through visual inspection.

Disadvantages of the MODE:

- Does not possess the desired algebraic property of the mean that allows further manipulations.
- Like the median, observations from different data sets have to be merged to obtain a new mode, whether group or ungrouped data are involved.

Problem 3.1

Suppose a company employs 20 people. The president of the company earns \$200,000, the vice president earns \$75,000, and 18 employees earn \$10,000 each. Is the mean the best number to choose to represent the "average" salary of the company? Nominal, ordinal

Problem 3.2

Suppose nine students obtain the following scores on an exam: 30, 35, 40, 40, 92, 92, 93, 98, 99

Is the median the best "average" to represent the set of scores?

Problem 3.3

Is the mode an appropriate "average" for the following test scores? 40, 42, 50, 62, 63, 65, 98, 98

Problem 3.4

The 20 meetings of a square dance club were attended by 26, 25, 28, 23, 25, 24, 24, 23, 26, 26, 28, 26, 24, 32, 25, 27, 24, 23, 24, and 22 of its members.

Find the mode, median, and mean.

Problem 3.5

If the mean annual salary paid to the top of three executives of a firm is \$96, 000, can one of them receive an annual salary of \$300, 000?

Problem 3.6

An instructor counts the final examination in a course four times as much as of the four one-hour examinations. What is the average grade of a student who received grades of 74, 80, 61, and 77 in the four one- hour examinations and 83 in the final examination?

Problem 3.7

In 1980 a college paid its 52 instructors a mean salary of \$13, 200, its 96 assistant professors a mean salary of \$15, 800, its 67 associate professors a mean salary of \$18, 900, and its 35 full professors a mean salary of \$23, 500.

What was the mean salary paid to all the teaching staff of this college?

❖ Measures of Dispersion: Range, Variance, and Standard Deviation

While mean and median tell you about the center of your observations, they say nothing about how data are scattered. Variability or dispersion measures the extent to which data are spread out. The measures of variability for data that we look at are: the range, the interquartile range, the variance, and the standard deviation.

➤ The Range

To measure the variability between extreme values (i.e. smallest and largest values) one uses the range. The range is the difference between the largest and smallest values of a distribution.

Example: 1

Find the range of each of the following samples:

Sample 1: 6, 18, 18, 18, 18, 18, 18, 18, 18, 18.

Sample 2: 6, 6, 6, 6, 6, 6, 18, 18, 18, 18, 18.

Sample 3: 6, 7, 9, 11, 12, 14, 15, 16, 17, 18.

Solution

Sample 1: $18 - 6 = 12$, **Sample 2:** $18 - 6 = 12$, **Sample 3:** $18 - 6 = 12$.

As you can see from this example, each sample has a range $18 - 6 = 12$ but the spread out of values is quite different in each case. In Sample 1, the spread is uniform whereas it is not in Sample 3. This is a disadvantage of this kind of measure. The range tells us nothing about the dispersion of the values between the extreme (smallest and largest) values. A better understanding is obtained by determining **quartiles**.

➤ Quartiles and Percentiles

Recall that if a set of data is arranged from smallest to largest, the middle value Q_2 (or the arithmetic mean of the two middle values) that divides the set of observations into two equal parts I_1 and I_2 is called the median of the distribution. That is, 50 percent of the observations are larger than the median and 50 percent are smaller.

Now, the median of I1 is denoted by Q1 (called the **lower quartile**) and that of I2 by Q3 (called the **upper quartile**). Thus, Q1, Q2 and Q3 divide the set of data into four equal parts. We call Q1, Q2 and Q3 the three quartiles of the distribution.

Example 1-3: Finding Quartiles

<u>Sales</u>	<u>Sorted Sales</u>		$(n+1)P/100$	
9	6			
6	9			
12	10			
10	12			
13	13	← First Quartile	$(20+1)25/100=5.25$	$13 + (.25)(1) = 13.25$
15	14			
16	14			
14	15			
14	16			
16	16	← Median	$(20+1)50/100=10.5$	$16 + (.5)(0) = 16$
17	16			
16	17			
24	17			
21	18			
22	18	← Third Quartile	$(20+1)75/100=15.75$	$18 + (.75)(1) = 18.75$
18	19			
19	20			
18	21			
20	22			
17	24			

In a similar manner, one could construct other measures of variation by considering percentiles rather than quartiles. For a whole number P, where $1 \leq P \leq 99$, the P^{th} percentile of a distribution is a value such that P% of the data fall at or below it. Thus, there are 99 percentiles. The percentiles locations are found using the formula

$$L_P = (n + 1) \cdot \frac{P}{100}$$

Thus, the median Q2 is the 50th percentile so that $P = 50$ and $L_{50} = \frac{n+1}{2}$

which is the median rank. Note that Q1 is the 25th percentile and Q3 is the 75th percentile.

An example will help to explain further.

Example 2

Listed below are the commissions earned, in dollars, last month by a sample of 15 brokers at Coca Cola Company, Ethiopia country director office.

2038	1758	1721	1637	2097
2047	2205	1787	2287	1940
2311	2054	2406	1471	1460

- (a) Rank the data from smallest to largest.
- (b) Find the median rank and then the median.
- (c) Find the quartiles Q1 and Q3.

Solution

(a) Arranging the data from smallest to largest we find

1460 1471 1637 1721 1758 1787 1940 2038 2047 2054 2097 2205 2287 2311 2406

(b) The median Q2 is the 50th percentile so that $P = 50$ and $L50 = \frac{15+1}{2} = 8$.

Thus, $Q2 = 2038$.

(c) Q1 is the 25th percentile so that $L25 = (15+1) \cdot \frac{25}{100} = 4$ and so $Q1 = 1721$.

Similarly, Q3 is the 75th percentile so that $P = 75$ and $L75 = (15+1) \cdot \frac{75}{100} = 12$. Thus, $Q3 = 2205$.

Percentiles are extensively used in such fields as educational testing. Undoubtedly some of you have had the experience of being told at what percentile you rated on a scholastic aptitude test.

Example 3.9

You took the English achievement test to obtain college credit in freshman English by examination.

If your score was in the 89th percentile, what does this mean?

Solution

This means 89% of the scores were at or below yours.

Remark 1

In Example 32.8, LP was found to be a whole number. That is not always the case. For example, if $n = 20$ observations then $L25 = (20+1) \cdot \frac{25}{100} = 5.25$.

In this case, to find Q1 we locate the fifth value in the ordered array and then move .25 of the distance between the fifth and sixth values and report that as the first quartile. Like the median, the quartile does not need to be one of the actual values in the data set.

To explain further, suppose a data set contained the six values: 91, 75, 61, 101, 43, and 104. We want to locate the first quartile. We order the values from smallest to largest: 43, 61, 75, 91, 101, and 104. The first quartile is located at

$$L25 = (6 + 1) \cdot \frac{25}{100} = 1.75$$

The position formula tells us that the first quartile is located between the first and the second value and that it is .75 of the distance between the first and the second values. The first value is 43 and the second is 61. So the distance between these two values is 18. To locate the first quartile, we need to

move .75 of the distance between the first and second values, so $.75(18) = 13.5$. To complete the procedure, we add 13.5 to the first value and report that the first quartile is 56.5.

➤ **Box-and-Whisker Plot**

A box-and-Whisker plot is a graphical display, based on quartiles, that helps us picture a set of data.

The steps for making such a box are as follows:

- Draw a vertical (or horizontal) scale to include the lowest and highest data values.
- To the right (or upper for the vertical scale) of the scale draw a box from Q1 to Q3.
- Include a solid line through the box at the median level.
- Draw solid lines, called Whiskers, from Q1 to the lowest value and from Q3 to the largest value.

An example will help to explain.

Example 3

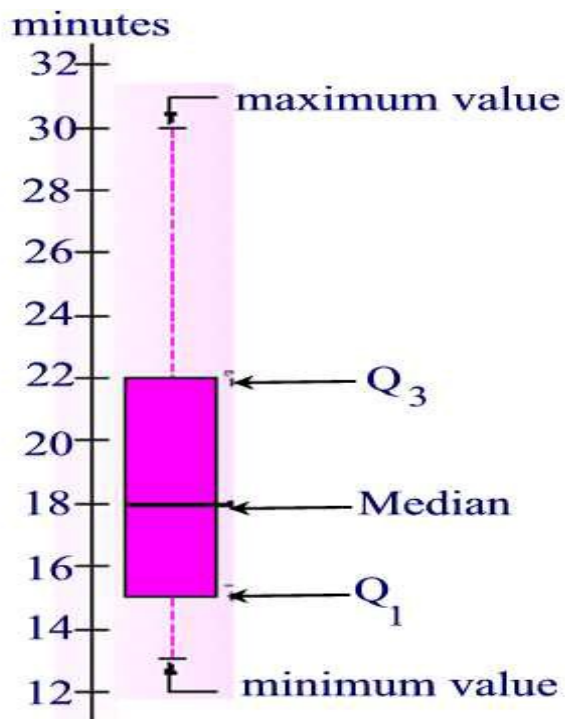
Alexanders Pizza offers free delivery of its pizza within 15 km. Alex, the owner, wants some information on the time it takes for delivery. How long does a typical delivery take? Within what range of times will most deliveries be completed? For a sample of 20 deliveries, he determined the following information:

Minimum value = 13 minutes, Q1 = 15 minutes

Median = 18 minutes, Q3 = 22 minutes and Maximum value = 30 minutes

Develop a box plot for the delivery times. What conclusions can you make about the delivery times?

Solution: The box plot is shown in Figure 3.2



The box plot shows that the middle 50 percent of the deliveries take between 15 minutes and 22 minutes. The distance between the ends of the box, 7 minutes, is the *inter-quartile range*. The inter-quartile range is the distance between the first and the third quartile. It tells us the spread of the middle half of the data.

The box plot also reveals that the distribution of delivery times is positively skewed since the median is not in the center of the box and the distance from the first quartile to the median is smaller than the distance from the median to the third quartile.

Practice Problems

Problem 3.9

The following table gives the average costs of a single-lens reflex camera:

800	650	300
430	560	470
640	830	400
280	800	410
360	600	310
370		

- Rank the data from smallest to largest.
- Find the quartiles Q_1 , Q_2 , and Q_3 .
- Make a box-and-whisker plot.

Problem 3.10

Mr. Eyha took a general aptitude test and scored in the 82nd percentile for aptitude in accounting. What percentage of the scores was at or below his score? What percentage was above?

Problem 3.11

At Center Hospital there is a concern about the high turnover of nurses. A survey was done to determine how long (in months) nurses had been in their current positions. The responses of 20 nurses were: 23 2 5 14 25 36 27 42 12 8 7 23 29 26 28 11 20 31 8 36

- (a) Rank the data.
- (b) Make a box-and-whisker plot of the data.
- (c) What are your conclusions from the plot?

➤ **Variance and Standard Deviation**

We have seen that a remedy for the deficiency of the range is the use of box and whisker plot. However, an even better measure of variability is the standard deviation. Unlike the range, the variance combines all the values in a data set to produce a measure of spread. The variance and the standard deviation are both measures of the spread of the distribution about the mean.

If μ is the population mean of set of data then the quantity $(x - \mu)$ is called the deviation from the mean. The variance of a data set is the arithmetic average of squared deviation from the mean. It is

denoted by s^2 and is given by the formula
$$s^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$$

Note that the variance is nonnegative, and it is zero only if all observations are the same.

The standard deviation is the square root of the variance:
$$s = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2}$$

The variance is the nicer of the two measures of spread from a mathematical point of view, but as you can see from the algebraic formula, the physical unit of the variance is the square of the physical unit of the data. For example, if our variable represents the weight of a person in pounds, the variance measures spread about the mean in squared pounds. On the other hand, standard deviation measures spread in the same physical unit as the original data, but because of the square root, is not as nice mathematically. Both measures of spread are useful.

A step by step approach to finding the standard deviation is:

1. Calculate the mean.
2. Subtract the mean from each observation.
3. Square each result.
4. Add these squares.
5. Divide this sum by the number of observations.

6. Take the positive square root.

The variance and standard deviation introduced above are for a population.

We define the sample variance by the formula

$$s^2 = \sum_{i=1}^n \left(\frac{x_i - \mu}{n} \right)^2 \quad \text{and the sample standard deviation by the formula}$$

$$= \frac{\sqrt{\sum_{i=1}^n (x_i - \mu)^2}}{n - 1}$$

The reason that for s we use n-1 instead of n is because usually a sample does not contain extreme values and we want s to be an estimate of σ therefore by using n - 1 we make s a little larger than if we divide by n.

Example 32.11

The owner of the Ches Tahoe restaurant is interested in how much people spend at the restaurant. He examines 10 randomly selected receipts for parties of four and writes down the following data.

44 50 38 96 42 47 40 39 46 50

(a) Find the arithmetic mean.

(b) Find the variance and the standard deviation.

Solution

(a) The arithmetic mean is the sum of the above values divided by 10, i.e., $\bar{x} = 49.2$

Below is the table for getting the standard deviation:

x	$x - 49.2$	$(x - 49.2)^2$
44	-5.2	27.04
50	0.8	0.64
38	11.2	125.44
96	46.8	2190.24
42	-7.2	51.84
47	-2.2	4.84
40	-9.2	84.64
39	-10.2	104.04
46	-3.2	10.24
50	0.8	0.64
Total		2600.4

Hence the variance is $s^2 = \frac{2600.4}{9}$

$= 289$ and the standard deviation is

$s = \sqrt{289} = 17$. What this means is that most of the patterns probably spend between $49.20 - 17 = \$32.20$ and $49.20 + 17 = \$66.20$.

Practice Problems

Problem 3.12: The following are the wind velocities reported at 6 P.M. on six consecutive days: 13, 8, 15, 11, 3 and 10. Find the range, sample mean, sample variance, and sample standard deviation.

Problem 3.13: An airline's records show that the flights between two cities arrive on the average 4.6 minutes late with a standard deviation of 1.4 minutes. At least what percentage of its flight between these two cities arrives anywhere between 1.8 minutes late and 7.4 minutes late?

Problem 3.14: One patient's blood pressure, measured daily over several weeks, averaged 182 with a standard deviation of 5.3, while that of another patient averaged 124 with a standard deviation of 9.4. Which patient's blood pressure is relatively more variable?

❖ Normal Distribution

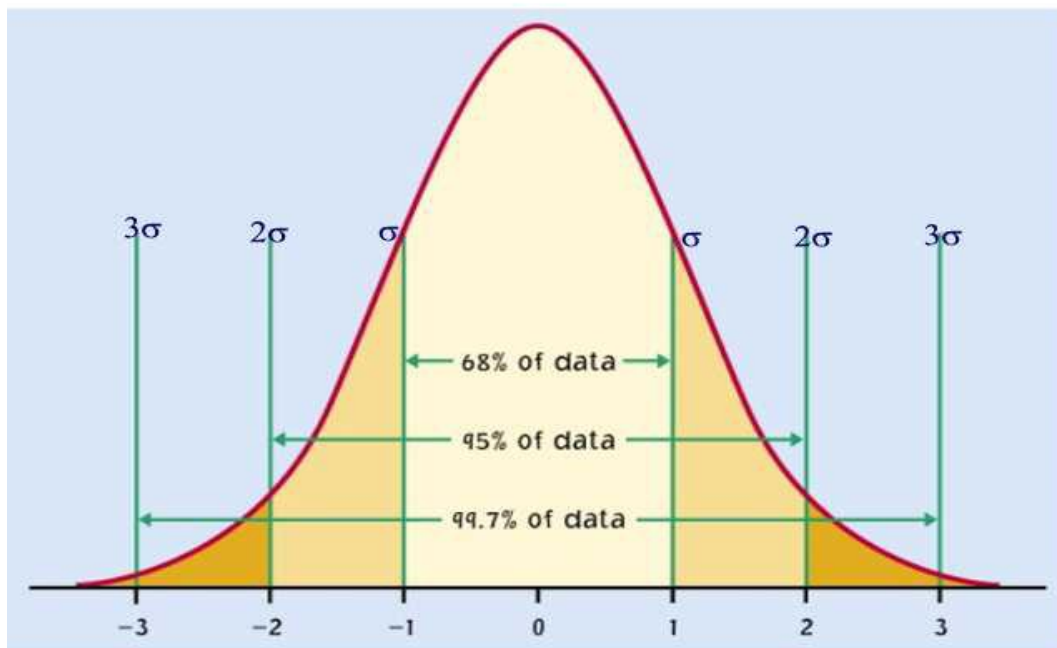
To better understand how standard deviations are used as measures of dispersion, we next consider normal distributions. The graph of a normal distribution is called a normal curve or a bell-shaped curve. The curve has the following properties:

1. The curve is bell-shaped with the highest point over the mean μ .
2. It is symmetrical about the line through μ .
3. The curve approaches the horizontal axis but never touches or crosses it.
4. The points where the curve changes concavity occur at $\mu + \sigma$ and $\mu - \sigma$.
5. The total area under the curve is assumed to be 1.(0.5 to the left of the mean and 0.5 to the right).

The data for a normal distribution are spread according to the following rules; (See Figure 32.2):

- About 68 percent of the data values will lie within one standard deviation of the mean.
- About 95 percent of the data values will lie within two standard deviations of the mean.
- About 99.7 percent of the data values will lie within three standard deviations of the mean.

This result is sometimes referred to as the empirical rule, because the given percentages are observed in practice.



Example 3.12

When a standardized test was scored, there was a mean of 500 and a standard deviation of 100. Suppose that 10,000 students took the test and their scores had a bell-shaped distribution.

- (a) How many scored between 400 and 600?
- (b) How many scored between 300 and 700?
- (c) How many scored between 200 and 800?

Solution.

- (a) Since one standard deviation on either side of the mean is from 400 to 600, about 68% of the scores fall in this interval. Thus, $0.68 \times 10,000 = 6800$ students scored between 400 and 600.
- (b) About 95% of 10,000 or 9500 students scored between 300 and 700.
- (c) About 99.7% of 10,000 or 9970 students scored between 200 and 800

By the discussion above we get information of the percentage of data within a certain number of standard deviations. However, if we want to find the location of a data value x from the mean we can use the so-called z -score given by the formula

$$z = \frac{x - \mu}{s}$$

Now, if the z -score is given then the raw data can be found by solving the equation $z = \frac{x - \mu}{s}$

for x to obtain, $x = \mu + zs$.

S

Note that from this formula we see that the value of z tells us how many standard deviation the corresponding value of x lies above (if $z > 0$) or below (if $z < 0$) the mean of its distribution.

Example 3.13

Scores on intelligence tests (IQs) are normally distributed in children. IQs from the Wechsler intelligence tests are known to have means of 100 and standard deviations of 15. In almost all the states in the United States (Pennsylvania and Nebraska are exceptions) children can be labeled as mentally retarded if their IQ falls to 70 points or below. What is the maximum z score one could obtain on an intelligence test and still be considered to be mentally retarded?

Solution

Applying the z -score formula we find $z = \frac{70-100}{15} = -2$.

Example 3.14

In a certain city the mean price of a quart of milk is 63 cents and the standard deviation is 8 cents. The average price of a package of bacon is \$1.80 and the standard deviation is 15 cents. If we pay \$0.89 for a quart of milk and \$2.19 for a package of bacon at a 24-hour convenience store, which is relatively more expensive?

Solution;

To answer this, we compute z -scores for each:

$$Z \text{ Milk} = \frac{0.89 - 0.63}{0.08} = 3.25$$

$$\text{And } z \text{ Bacon} = \frac{2.19 - 1.80}{0.15} = 2.60$$

Our z -scores show us that we are overpaying quite a bit more for the milk than we are for the bacon.

Example 3.15

Graduate Record Examination (GRE) scores have means equal to 500 and standard deviations of 100. If a person receives a z -score on the GRE of 1.45, what would their raw score be?

Solution;

Using the formula $x = \mu + z\sigma$ we find $x = 500 + 1.45(100) = 645$.